Q68 UPDATE: THE S_{16} CENSUS IS CLOSED (114/296/0), AND "(II) ALWAYS KILLS" IS FALSE

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Corresponds to research-repo commit `98a9d52` (`grandpa-rick/rick-research`; record `proofs/2026-11-08-q68-lemmaI-false-s16-closed.md`, data `proofs/prove283/`).
Follows: `notes/2026-11-07-q68-s16-correction.tex` (PDF `0026ffa`, built against `498fdd1`), §3 census and §4 item 3.
Registry `proofs/registry/seedQ2-kl-cone.json`.

In one paragraph. The 11-07 correction note left 7 S_{16} candidate pairs UNDECIDED and stated, graded hunch, that "(ii) always kills" (56/56 decided instances had $\mu(y,z)=0$), adding that if it held, 5 of the 7 would be NO. Both open ends are now settled. The 5 pairs of shape $(5,5,3,3)$ are NO, by layer certificates whose kills are certified $\mu(y,z)=0$ and do not use the hunch. The 2 pairs of shape $(6,4,3,3)$ are **TWIN**. Their decisive (ii)-element has $\mu(y,z)=1$, so **"(ii) always kills" is false**. An intermediate x-free reduction ("Lemma I") is also false, with explicit counterexamples. **Grades:** every new μ value here is computed from an exact KL computation on the Bruhat interval, and the two decisive ones were replayed by a second, from-scratch implementation. Nothing here is proved.

S_{16} census: 410 pairs, 114 **TWIN** / 296 **NO** / 0 **undecided**, §1	computed
"(ii) always kills" (11-07 correction §4.3)	dead-end (refuted, computed)
Lemma I (x-free reduction, below)	dead-end (refuted, computed)
Layer Lemma $P_{v'u,vu}=P_{v',v}$	folklore [BB, Thm 6], [Po, L. 2.6]

Conventions are as in the 11-07 notes: one-line permutations; $L(w)=\mathrm{Des}P(w)$; left cells $\{Q=\mathrm{const}\}$; and the cell $Q_0=[[1..6],[7..10],[11..13],[14..16]]$. A *candidate pair* (x,z) lies in one cell, with $L(x)=A$ and $\varnothing\neq L(z)\subseteq \mathrm{int}\,A$. Its *(i)/(ii)-elements* are as in the correction note, §1. By Lemma A, the pair is a twin iff $\mu(y,z)\geq\mu(y,x)$ on (i)∪(ii).

1. The census is closed

The five $(5,5,3,3)$ *pairs: NO.* A missing $\mu(y,z)$ (3 (ii)-elements, 2 (i)-elements) is carried by an explicit right-star path (Lemma S) to a pair in which 16 (or 1) sits in the same row of both tableaux. The *Layer Lemma* then gives $P_{v'u,vu}=P_{v',v}$ for $u\in {}^KW$ and $v,v'\in W_K$. This deletes the letter and lands on a Bruhat-incomparable pair in S_{15}, so $\mu=0$. The certificates were replayed by fresh code. The Layer Lemma is folklore: Billey–Braden [BB, Thm 6] call it "well-known" and cite Polo [Po, Lemma 2.6]. I read Billey–Braden first-hand; the Polo locator is second-hand.

The two $(6,4,3,3)$ *pairs: TWIN* (computed).

pair	x; z	$\ell(x),\ell(z)$
353	$[5,9,10,11,15,16,1,4,8,14,3,7,13,2,6,12]$; $[8,9,10,14,15,16,4,11,12,13,5,6,7,1,2,3]$	63, 81
364	$[1,5,6,11,15,16,4,7,10,14,3,9,13,2,8,12]$; $[4,5,10,14,15,16,6,11,12,13,7,8,9,1,2,3]$	53, 73

For 353, $A=\{2,3,4,6,7,8,12,13,14\}$ and $L(z)=\{3,7,13\}$. For 364, $A=\{2,3,4,8,9,10,12,13,14\}$ and $L(z)=\{3,9,13\}$.

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- *Elements certified by star paths.* All (i)∪(ii) elements have $\mu(y,z) = 1$ by star certificates, except one per pair. Pair 353 has 5 of 6 certified; pair 364 has 6 of 7.
- *The decisive elements.* For 353 it is the (ii)-element $y_0 = [1,5,9,11,15,16,4,8,10,14,3,7,13,2,6,12]$, with $\ell = 58$ and $\mathrm{Des}P_{y_0} = A \cup \{10\}$. For 364 it is $y_1 = [1,5,7,11,15,16,4,6,10,14,3,9,13,2,8,12]$, with $\ell = 54$. This y_1 is both the (i)-element at $s = 6$ and the (ii)-element, and $\mathrm{Des}P_{y_1} = A \cup \{6\}$.
- *Moving each decisive pair.* A right-star path of 54 steps (resp. 19) moves (y_0, z) (resp. (y_1, z)) to a short pair with the same (P, P). On that pair the full KL polynomial was computed:

	ℓ (rep.)	$\lvert[\cdot,\cdot]\rvert$ (rep.)	$P_{x,z}$	μ
rep. of (y_0, z)	33, 52	894,344	$1 + 12q + 62q^2 + 181q^3 + 328q^4 + 381q^5 + 279q^6 + 119q^7 + 26q^8 + q^9$	1
rep. of (y_1, z)	35, 54	1,049,576	$1 + 13q + 72q^2 + 220q^3 + 403q^4 + 455q^5 + 314q^6 + 123q^7 + 26q^8 + q^9$	1

The (y_0, z) representative is $x' = [5,4,3,1,9,8,7,2,15,14,11,6,13,10,12,16]$, $z' = [14,8,4,1,15,9,5,2,16,10,6,3,11,7,12,13]$.

- *How P was computed.* Engine 1 is Dyer's Bruhat-graph path formula for $\tilde{R}$, with lex reflection order, inverted top-down on the interval. It uses the double-coset shortcut $P_{y,z} = P_{\max(W_{L(z)}yW_{R(z)}),z}$.
- *Independent replay of (y_0, z).* Engine 2 was written from scratch without reading engine 1. It uses Deodhar's parabolic KL recursion on W^J, $J = R(z')$, over 22,208 cosets. It reproduces $\lvert[x', z']\rvert$ and P exactly.
- *Validation.* Both engines agree with brute-force KL on S_5 and S_6 (0 mismatches). That is the limit of brute-force validation.
- *Caveats.* (y_1, z) is the evacuation mirror of (y_0, z) and was not independently replayed. Engine 2 ran on the representative printed above. The certificate's path endpoint is a second representative with the same (P_x, P_z), on which engine 1 gives the identical P.
- *Certificate replay.* The certificates, including the path replay and a literal endpoint match against the KL input, were replayed by the census verifier. Six deliberately tampered certificates were all rejected: verdict flips, a truncated path, swapped instance data, a dropped y, and an edited μ.

So the census goes from 112/291/7 to **114/296/0**. The shape count is unchanged (17/21 shapes carry a twin), so the 11-07 correction stands as sent.

2. "(ii) always kills" is false

Pairs 352 and 365 are parity-OK candidate pairs in Q_0 with the same z as 353 and 364 respectively. Their (ii)-elements are *the same* y_0 and y_1 (for 352, $y_0 = s_{10}x$ with $\ell(x) = 57$). Hence $\mu(y,z) = 1$ at a (ii)-element. Pairs 353 and 364 are also witnesses: a pair with a non-empty (ii)-set can be a twin only if its (ii)-element does *not* kill. Pairs 352 and 365 remain NO for other reasons, since other (i)-elements have certified $\mu(y,z) = 0$.

3. Lemma I is false

In unsent work (PROVE 282) I proved two things. First, "(ii) always kills" holds whenever $\ell(x) \not\equiv \ell(z)$; this is a parity argument. Second, the parity-OK case reduces to an x-free statement:

Lemma I (conjectured in PROVE 282, unsent). If $x \sim_L z$, $L(z) \neq \varnothing$ and $N[L(z)] \subseteq L(x) \neq S$, then $\mu(x,z) = 0$.

Here $N[\cdot]$ is the closed Dynkin neighbourhood. Lemma I fails: the pair (y_0, z) above already satisfies its hypotheses. A smaller, *tight* counterexample ($L(x) = N[L(z)]$) has shape $(4, 4, 4, 3, 1)$:

$$x = [7, 3, 2, 11, 10, 6, 5, 1, 15, 9, 8, 4, 16, 14, 13, 12],$$
$$z = [10, 6, 2, 14, 11, 7, 3, 1, 15, 12, 8, 4, 16, 13, 9, 5].$$

Here $\ell = 42, 55$, $L(x) = \{1, 2, 4, 5, 6, 8, 9, 10, 12, 13, 14\}$ and $L(z) = \{1, 5, 9, 13\}$. We have $R(x) = R(z)$, and $|[x, z]| = 20{,}160$. The KL polynomial is

$$P_{x,z} = 1 + 8q + 24q^2 + 35q^3 + 26q^4 + 9q^5 + q^6, \qquad \mu(x, z) = 1.$$

Engine 1 gives this value. A from-scratch R-polynomial recursion over the full interval gives the identical P (computed, independently replayed). A further tight instance, $(5, 5, 4, 2)$ with gap 15, also has $\mu = 1$ (engine 1 only).

Owner note. I found no arc-exclusion rule of this kind in Stembridge's W-graph papers (2008/2010/2012), in Chmutov (1307.8354), or in Chmutov–Lewis–Pylyavskyy and Chmutov–Pylyavskyy–Yudovina. The nearest framework is Nguyen (1807.07457, Prop. 9.5). That is now moot. My earlier guess that Lemma I "is" a Stembridge arc rule is withdrawn.

4. What this changes

- Nothing in the 11-07 notes changes, apart from the hunch in the correction note's §4 item 3, which is now refuted. No census verdict ever used "(ii) always kills" or Lemma I.
- In type A, twin existence is now known pair-by-pair through $n = 16$. Even with (ii)-elements present, it is *not* decided by descent data plus parity.

References

[BB] S. Billey, T. Braden, *Lower bounds for Kazhdan–Lusztig polynomials from patterns*, Transform. Groups 8 (2003); arXiv:math/0202252, Thm. 6.

[Po] P. Polo, *Construction of arbitrary Kazhdan–Lusztig polynomials in symmetric groups*, Represent. Theory 3 (1999) 90–104, Lemma 2.6 (cited via [BB]; not opened).